Equations Used by the Multi-Ellipsoid Boundary-Integral Solver

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Abstract

This note records the mathematical model and discrete equations implemented in the current multi-ellipsoid solver. The fluid is inviscid, incompressible and irrotational, so the hydrodynamic coupling is obtained from a Neumann boundary-integral solve for the velocity potential. The preferred multibody time step is an impulse/Hamiltonian endpoint formulation: fluid impulse and total kinetic energy are treated as state functions of the rigid-body configuration and velocities, and the endpoint velocities are chosen to satisfy the global linear impulse, angular impulse and energy constraints.

Contents

1 Rigid bodies and notation

The system consists of N rigid ellipsoids immersed in an unbounded ideal fluid of density ρ_f. Body b has centre $\boldsymbol{X}_b$, orientation quaternion $\mathbf{q}_b$, semi-axes (a_b, b_b, c_b), solid density $\rho_{s,b}$ and volume

$$V_b = \frac{4\pi}{3} a_b b_b c_b. \tag{1}$$

The solid mass and body-frame inertia tensor are

$$m_b = \rho_{s,b} V_b, \qquad \boldsymbol{I}_{s,b} = \frac{m_b}{5} \begin{bmatrix} b_b^2 + c_b^2 & 0 & 0 \\ 0 & a_b^2 + c_b^2 & 0 \\ 0 & 0 & a_b^2 + b_b^2 \end{bmatrix}. \tag{2}$$

The lab-frame linear velocity is $\boldsymbol{U}_b = \dot{\boldsymbol{X}}_b$. The lab angular velocity is represented as the pure quaternion $(0, \boldsymbol{\Omega}_b)$; body-frame vectors are obtained by

$$(0, \boldsymbol{\omega}_b) = \mathbf{q}_b^{-1}(0, \boldsymbol{\Omega}_b)\mathbf{q}_b. \tag{3}$$

The solid angular momentum in the lab frame is therefore

$$\boldsymbol{h}_{s,b} = \mathbf{q}_b \left(0, \boldsymbol{I}_{s,b}\boldsymbol{\omega}_b\right) \mathbf{q}_b^{-1}. \tag{4}$$

The solid kinetic energy is

$$T_{s,b} = \frac{1}{2} m_b \|\boldsymbol{U}_b\|^2 + \frac{1}{2}\boldsymbol{\omega}_b^\mathsf{T} \boldsymbol{I}_{s,b}\boldsymbol{\omega}_b. \tag{5}$$

2 Single-ellipsoid added-mass tensors

For an isolated ellipsoid the solver also computes the classical added-mass reference tensors from the shape functions

$$\alpha_i = abc \int_0^\lambda \frac{\mathrm{d}s}{(a_i^2 + s)\sqrt{(a^2+s)(b^2+s)(c^2+s)}}, \qquad (a_1, a_2, a_3) = (a, b, c). \tag{6}$$

In the code this integral is evaluated with a large finite upper limit $\lambda = 10^4$. Writing $(\alpha_1, \alpha_2, \alpha_3) = (\alpha, \beta, \gamma)$, the translational added-mass tensor is

$$\boldsymbol{M}_f = \rho_f V \operatorname{diag}\left(\frac{\alpha}{2-\alpha}, \frac{\beta}{2-\beta}, \frac{\gamma}{2-\gamma}\right). \tag{7}$$

The rotational fluid inertia reference used for diagnostics is

$$\boldsymbol{I}_f = \frac{\rho_f V}{5} \operatorname{diag}(e_1, e_2, e_3), \tag{8}$$

where

$$e_1 = \frac{(b^2 - c^2)^2(\gamma - \beta)}{2(b^2 - c^2) + (\beta - \gamma)(b^2 + c^2)}, \tag{9}$$

$$e_2 = \frac{(a^2 - c^2)^2(\gamma - \alpha)}{2(a^2 - c^2) + (\alpha - \gamma)(a^2 + c^2)}, \tag{10}$$

$$e_3 = \frac{(a^2 - b^2)^2(\beta - \alpha)}{2(a^2 - b^2) + (\alpha - \beta)(a^2 + b^2)}. \tag{11}$$

For a single isolated body the reference Hamiltonian is then

$$H_{\text{iso}} = \frac{1}{2}\boldsymbol{U}^\mathsf{T}(\boldsymbol{M}_s + \boldsymbol{M}_f)\boldsymbol{U} + \frac{1}{2}\boldsymbol{\omega}^\mathsf{T}(\boldsymbol{I}_s + \boldsymbol{I}_f)\boldsymbol{\omega}. \tag{12}$$

3 Potential-flow boundary integral

The fluid velocity is $\boldsymbol{u} = \nabla\phi$ and the potential satisfies

$$\nabla^2\phi = 0 \quad \text{in the exterior fluid domain}, \qquad \nabla\phi(\boldsymbol{x}) \to 0 \quad \text{as } \|\boldsymbol{x}\| \to \infty. \tag{13}$$

On the surface S_b of body b, the no-penetration condition gives the Neumann data

$$\frac{\partial\phi}{\partial n}(\boldsymbol{x}) = \boldsymbol{n}(\boldsymbol{x}) \cdot \left[\boldsymbol{U}_b + \boldsymbol{\Omega}_b \times (\boldsymbol{x} - \boldsymbol{X}_b)\right], \qquad \boldsymbol{x} \in S_b. \tag{14}$$

The free-space Green function and its gradient are

$$G(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{4\pi\|\boldsymbol{x}-\boldsymbol{y}\|}, \qquad \nabla_{\boldsymbol{x}} G(\boldsymbol{x},\boldsymbol{y}) = -\frac{\boldsymbol{x}-\boldsymbol{y}}{4\pi\|\boldsymbol{x}-\boldsymbol{y}\|^3}. \tag{15}$$

The direct boundary-integral equation is assembled in the form

$$c(\boldsymbol{x})\phi(\boldsymbol{x}) + \int_S \phi(\boldsymbol{y})\frac{\partial G(\boldsymbol{x},\boldsymbol{y})}{\partial n_{\boldsymbol{y}}}\,\mathrm{d}S_{\boldsymbol{y}} = \int_S G(\boldsymbol{x},\boldsymbol{y})\frac{\partial \phi}{\partial n}(\boldsymbol{y})\,\mathrm{d}S_{\boldsymbol{y}}, \qquad \boldsymbol{x}\in S. \tag{16}$$

After discretisation on quadratic triangular elements this becomes the dense linear system

$$\boldsymbol{A}(\boldsymbol{X},\mathbf{q})\,\boldsymbol{\phi} = \boldsymbol{b}(\boldsymbol{X},\mathbf{q},\boldsymbol{U},\boldsymbol{\Omega}), \tag{17}$$

with $\boldsymbol{\phi}$ the nodal boundary potential. Matrix entries are generated by double-layer quadrature and the right-hand side by the single-layer integral of the imposed Neumann data. In the current multibody solve all bodies share one combined surface mesh, so cross-body hydrodynamic interactions are contained in the off-diagonal blocks of $\boldsymbol{A}$.

4 Surface state functions

Once (17) has been solved, the fluid kinetic energy is computed by Green's identity as the surface integral

$$T_f = -\frac{\rho_f}{2}\int_S \phi\,\frac{\partial\phi}{\partial n}\,\mathrm{d}S. \tag{18}$$

The minus sign follows the outward-normal convention of the exterior fluid problem used by the implementation.

The per-body Lamb impulse and angular Lamb impulse are

$$\boldsymbol{L}_b = \rho_f \int_{S_b} \phi\,\boldsymbol{n}\,\mathrm{d}S, \tag{19}$$

$$\boldsymbol{\Lambda}_b = \rho_f \int_{S_b} \phi\,(\boldsymbol{x}-\boldsymbol{X}_b)\times\boldsymbol{n}\,\mathrm{d}S. \tag{20}$$

These are state functions of the instantaneous configuration and generalized velocity. They are central to the impulse schemes because they do not require a history-dependent approximation to $\partial\phi/\partial t$.

For the pressure-force option, the unsteady potential term gives

$$\boldsymbol{F}_b^{(\dot\phi)} = \rho_f \int_{S_b} \dot\phi\,\boldsymbol{n}\,\mathrm{d}S, \tag{21}$$

$$\boldsymbol{N}_b^{(\dot\phi)} = \rho_f \int_{S_b} \dot\phi\,(\boldsymbol{x}-\boldsymbol{X}_b)\times\boldsymbol{n}\,\mathrm{d}S. \tag{22}$$

When the surface is rotating, the full time derivative of the impulse also contains transport terms. The implemented pressure-mode correction is

$$\boldsymbol{F}_b = \boldsymbol{F}_b^{(\dot\phi)} + \boldsymbol{\Omega}_b\times\boldsymbol{L}_b, \tag{23}$$

$$\boldsymbol{N}_b = \boldsymbol{N}_b^{(\dot\phi)} + \boldsymbol{\Omega}_b\times\boldsymbol{\Lambda}_b. \tag{24}$$

The missing angular version of (24) was the source of the old single-body impulse energy error.

5 Conserved impulse and Hamiltonian

The solver monitors and, in the Hamiltonian endpoint scheme, enforces the total linear and angular conserved impulses

$$\boldsymbol{P} = \sum_{b=1}^{N} (m_b \boldsymbol{U}_b - \boldsymbol{L}_b), \tag{25}$$

$$\boldsymbol{H} = \sum_{b=1}^{N} [\boldsymbol{X}_b \times (m_b \boldsymbol{U}_b - \boldsymbol{L}_b) + \boldsymbol{h}_{s,b} - \boldsymbol{\Lambda}_b]. \tag{26}$$

The total kinetic energy used by the energy-conserving schemes is

$$H(\boldsymbol{X}, \mathbf{q}, \boldsymbol{U}, \boldsymbol{\Omega}) = \sum_{b=1}^{N} T_{s,b} + T_f. \tag{27}$$

Equations (25)–(27) are evaluated from the same BEM solve, so all added-mass and body–body interaction terms are included implicitly.

6 Impulse-difference time step

Let h be the time step and let $(\boldsymbol{L}_b^n, \boldsymbol{\Lambda}_b^n)$ denote the BEM impulse at the start of the step. The implicit impulse-difference update solves for the endpoint velocity and a step-averaged torque. The translational equation is

$$m_b \left(\boldsymbol{U}_b^{n+1} - \boldsymbol{U}_b^n\right) = \boldsymbol{L}_b^{n+1} - \boldsymbol{L}_b^n + h\, \boldsymbol{G}_{X,b}, \tag{28}$$

where $\boldsymbol{G}_X$ is normally zero and is only nonzero when the experimental finite difference gradient of the fluid kinetic energy is enabled. Positions use the midpoint velocity,

$$\boldsymbol{X}_b^{n+1} = \boldsymbol{X}_b^n + \frac{h}{2}\left(\boldsymbol{U}_b^n + \boldsymbol{U}_b^{n+1}\right). \tag{29}$$

The consistent torque uses the angular impulse difference about the translating body centre and the reference-point transport correction

$$\boldsymbol{N}_b = \frac{\boldsymbol{\Lambda}_b^{n+1} - \boldsymbol{\Lambda}_b^n}{h} - \bar{\boldsymbol{U}}_b \times \bar{\boldsymbol{p}}_b + \boldsymbol{G}_{q,b}, \qquad \bar{\boldsymbol{p}}_b = \frac{1}{2}\left[m_b \boldsymbol{U}_b^n - \boldsymbol{L}_b^n + m_b \boldsymbol{U}_b^{n+1} - \boldsymbol{L}_b^{n+1}\right]. \tag{30}$$

The angular velocity is advanced with a Lie-group implicit midpoint rule in the body frame:

$$\boldsymbol{\omega}_{b,m} = \boldsymbol{\omega}_b^n + \frac{h}{2} \boldsymbol{I}_{s,b}^{-1} \left[\boldsymbol{N}_{b,m} - \boldsymbol{\omega}_{b,m} \times (\boldsymbol{I}_{s,b} \boldsymbol{\omega}_{b,m})\right], \tag{31}$$

$$\mathbf{q}_b^{n+1} = \exp\left(\frac{h}{2}(0, \boldsymbol{\Omega}_{b,m})\right) \mathbf{q}_b^n, \tag{32}$$

$$\boldsymbol{\omega}_b^{n+1} = 2\boldsymbol{\omega}_{b,m} - \boldsymbol{\omega}_b^n. \tag{33}$$

Here $\boldsymbol{N}_{b,m}$ is the applied torque rotated into the midpoint body frame. The nonlinear fixed point stacks the unknown endpoint velocities and torques, and Anderson acceleration is applied to that stacked residual.

7 Hamiltonian midpoint endpoint solve

The most conservative multibody option solves directly for generalized endpoint data. Define the generalized velocity vector

$$\boldsymbol{z} = [\boldsymbol{U}_1, \ldots, \boldsymbol{U}_N, \boldsymbol{\Omega}_1, \ldots, \boldsymbol{\Omega}_N]^\mathsf{T} \in \mathbb{R}^{6N}. \tag{34}$$

For a midpoint-velocity version of the method, $\boldsymbol{z}$ defines the endpoint configuration by

$$\boldsymbol{X}_b^{n+1}(\boldsymbol{z}) = \boldsymbol{X}_b^n + h\,\boldsymbol{U}_{b,m}, \tag{35}$$

$$\mathbf{q}_b^{n+1}(\boldsymbol{z}) = \exp\left(\frac{h}{2}(0, \boldsymbol{\Omega}_{b,m})\right)\mathbf{q}_b^n. \tag{36}$$

The midpoint configuration used for the BEM solve is reconstructed from the start and endpoint states:

$$\boldsymbol{X}_{b,m} = \frac{1}{2}\left(\boldsymbol{X}_b^n + \boldsymbol{X}_b^{n+1}\right), \tag{37}$$

$$\boldsymbol{\Omega}_{b,m} = \frac{1}{h}\log\left(\mathbf{q}_b^{n+1}(\mathbf{q}_b^n)^{-1}\right), \tag{38}$$

$$\mathbf{q}_{b,m} = \exp\left(\frac{h}{4}(0, \boldsymbol{\Omega}_{b,m})\right)\mathbf{q}_b^n. \tag{39}$$

The endpoint/coupled variant instead evaluates the conservation residual at the endpoint state generated by $\boldsymbol{z}$.

Let

$$\boldsymbol{C}(\boldsymbol{z}) = \begin{bmatrix} \boldsymbol{P}(\boldsymbol{z}) \\ \boldsymbol{H}(\boldsymbol{z}) \end{bmatrix}, \qquad \boldsymbol{C}^\star = \boldsymbol{C}(\boldsymbol{z}^n), \qquad H^\star = H(\boldsymbol{z}^0), \tag{40}$$

where $H^\star$ is the initial total kinetic energy. The seven-component nonlinear residual used by the coupled endpoint solve is

$$\boldsymbol{R}(\boldsymbol{z}) = \begin{bmatrix} (\boldsymbol{P}(\boldsymbol{z}) - \boldsymbol{P}^\star)/s_P \\ (\boldsymbol{H}(\boldsymbol{z}) - \boldsymbol{H}^\star)/s_H \\ (H(\boldsymbol{z}) - H^\star)/s_E \end{bmatrix}, \tag{41}$$

with scales

$$s_P = \max(\|\boldsymbol{P}^\star\|, 1), \qquad s_H = \max(\|\boldsymbol{H}^\star\|, 1), \qquad s_E = \max(|H^\star|, 1). \tag{42}$$

The code solves (41) by finite-difference Newton steps, with optional Broyden rank-one Jacobian updates and a trust-style cap on the generalized velocity increment.

8 Energy and impulse projection

The projection used by the Hamiltonian experiments is algebraic. At fixed configuration the conserved impulse is affine in generalized velocity,

$$\boldsymbol{C}(\boldsymbol{z}) = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{c}_0, \tag{43}$$

where $\boldsymbol{A}$ is obtained by evaluating the BEM state function on basis velocity vectors. A first correction projects the current velocity $\boldsymbol{z}_c$ onto the linear impulse constraint:

$$\boldsymbol{z}_P = \boldsymbol{z}_c + \boldsymbol{A}^\mathsf{T}(\boldsymbol{A}\boldsymbol{A}^\mathsf{T})^{-1}\left(\boldsymbol{C}^\star - \boldsymbol{C}(\boldsymbol{z}_c)\right). \tag{44}$$

A minimum-energy particular solution satisfying the same affine constraint is computed from the sampled energy quadratic

$$H(\boldsymbol{z}) = H(\boldsymbol{0}) + \frac{1}{2}\boldsymbol{z}^\mathsf{T}\boldsymbol{K}\boldsymbol{z}, \tag{45}$$

as

$$\boldsymbol{z}_0 = \boldsymbol{K}^{-1}\boldsymbol{A}^{\mathsf{T}}\left(\boldsymbol{A}\boldsymbol{K}^{-1}\boldsymbol{A}^{\mathsf{T}}\right)^{-1}(\boldsymbol{C}^{\star} - \boldsymbol{c}_0)\,. \tag{46}$$

The remaining correction is restricted to the null direction

$$\boldsymbol{d} = \boldsymbol{z}_P - \boldsymbol{z}_0, \qquad \boldsymbol{z}(\alpha) = \boldsymbol{z}_0 + \alpha\boldsymbol{d}. \tag{47}$$

The scalar α is chosen from

$$H(\boldsymbol{z}_0 + \alpha\boldsymbol{d}) = H^{\star}. \tag{48}$$

In implementation the coefficients of this quadratic are evaluated by three BEM energy calls:

$$q_a = \frac{1}{2}\left[H(\boldsymbol{z}_P) + H(\boldsymbol{z}_0 - \boldsymbol{d})\right] - H(\boldsymbol{z}_0), \tag{49}$$

$$q_b = \frac{1}{2}\left[H(\boldsymbol{z}_P) - H(\boldsymbol{z}_0 - \boldsymbol{d})\right], \tag{50}$$

$$q_c = H(\boldsymbol{z}_0) - H^{\star}, \tag{51}$$

and the root closest to $\alpha = 1$ is selected. If the energy floor $H(\boldsymbol{z}_0)$ already exceeds the target, the exact energy projection is infeasible and the code falls back to $\boldsymbol{z}_0$ while reporting the floor error.

9 Diagnostics written to output

The CSV output writes the total energy components

$$H, \qquad T_f, \qquad \sum_b T_{s,b}, \tag{52}$$

and, for each body, the diagnostic vectors

$$\boldsymbol{p}_{\mathrm{con},b} = m_b\boldsymbol{U}_b - \boldsymbol{L}_b, \tag{53}$$

$$\boldsymbol{h}_{\mathrm{con},b} = \boldsymbol{X}_b \times \boldsymbol{p}_{\mathrm{con},b} + \boldsymbol{h}_{s,b} - \boldsymbol{\Lambda}_b. \tag{54}$$

For a closed multibody system the sums of these vectors are the conserved quantities in (25) and (26). The study dashboards report kinetic-energy drift, separation, trajectories, orientation markers, and relative drift in the summed conserved linear and angular impulses.

10 Remarks on physical interpretation

The BEM solve makes the fluid energy and impulse depend on the entire multibody configuration, not on a pairwise approximation. Low-density bodies therefore have small solid inertia relative to the configuration-dependent fluid inertia, which is precisely the regime in which explicit pressure-force updates drift or destabilise. The impulse/Hamiltonian schemes instead enforce the global invariants (25)–(27) at the endpoint level, making the energy behaviour much closer to the conservative ideal-fluid dynamics.